Supplemental Material

Table 1. *Claim phrases of the original authors in their discussion, where their hypothesis was supported.*

Term	Frequency	Term	Frequency
supported our hypothesis	5	our first hypothesis was accepted	1
as hypothesized	2	our hypotheses were accepted	1
confirmed our hypothesis	2	our hypothesis [...] was confirmed	1
in agreement with our hypothesis	2	our initial hypothesis was met	1
in support of our hypothesis	2	support our first hypothesis	1
our first hypothesis was confirmed	2	support our initial hypothesis	1
support the hypothesis	2	support the hypotheses	1
agreeing with our first hypothesis	1	support the study hypothesis	1
as expected	1	support this hypothesis	1
as per our original hypothesis	1	supported the authors' initial hypothesis	1
confirm the hypothesis	1	supported the first hypothesis	1
confirmed	1	supported the hypotheses	1
confirmed our initial hypotheses	1	supported the hypothesis	1
confirmed the initial hypothesis	1	supports our first hypothesis	1
consistent with our first experimental hypothesis	1	supports our primary hypothesis	1
consistent with our first hypothesis	1	supports the hypothesis	1
consistent with our hypothesis	1	the first hypothesis is supported	1
corroborating the hypothesis	1	the hypothesis [...] was supported.	1

Term	Frequency	Term	Frequency
fulfilling our hypothesis	1	the hypothesis of this investigation was confirmed	1
hypothesis 1 can be confirmed	1	we accepted the hypothesis	1
in accordance to our original hypothesis	1	we first hypothesized that [...] which was supported	1
in accordance with our hypothesis	1	which was consistent with the hypothesis	1
in line with our hypotheses	1		
in line with our hypothesis	1		
in line with our initial expectations	1		
in support of hypothesis 1	1		
in support of the first hypothesis	1		
in support of the hypothesis	1		
meet our hypotheses	1		

Table 2. *Claim phrases of the original authors in their discussion, where their hypothesis was not supported.*

Term	Frequency	Term	Frequency
in contrast to our hypothesis	5	in contrast to the main research hypothesis	1
contrary to our hypothesis	4	not consistent with a priori expectation because we hypothesized that	1
in contrast to our initial hypothesis	2	oppose our hypothesis	1
contradict our hypothesis	1	our claim [...] is not supported	1
contradict those predicted	1	our hypotheses [...] were not supported	1
contrary to our first hypothesis	1	our hypothesis [...] was not supported	1
contrary to our hypotheses	1	our hypothesis was rejected	1
contrary to our initial hypothesis	1	our main hypotheses were not supported	1
contrary to our original hypothesis	1	refuting the initial hypothesis	1
contrary to the tested hypotheses	1	rejects our hypothesis	1
did not support our hypothesis	1	the first and second hypotheses were not supported	1
did not support our initial hypothesis	1	the hypothesis [...] was not confirmed	1

Term	Frequency	Term	Frequency
did not support the hypotheses	1	the hypothesis [...] was not confirmed.	1
disagree with our hypothesis	1	the hypothesis [...] was not supported	1
in contradiction of our first hypothesis	1	this hypothesis was rejected	1

Table 3. *Claim phrases of the original authors in their discussion, where support for their hypothesis was "mixed".*

Term	Frequency	Term	Frequency
in partial agreement with our initial hypothesis	1	partially support our first hypothesis	1
in partial support of our hypothesis	1	partially supporting our first hypothesis	1
our first hypothesis was partially supported	1	partially supports our hypotheses	1
our hypotheses were partially supported	1	partly in accordance with our hypothesis	1
our hypotheses were partially true	1	partly in agreement with our first hypothesis	1
our hypothesis was not completely supported	1	partly support our first hypothesis	1
our hypothesis was partially supported	1	partly supporting this study's hypotheses	1
our initial hypothesis was partially confirmed	1	support part of this hypothesis	1
partially confirm our hypothesis	1	supported in part our hypotheses	1
partially confirm the hypothesis	1	the first hypothesis was partially supported	1
partially confirmed our initial hypothesis	1	the hypotheses of the study were partially confirmed	1
partially corroborate the initial hypothesis	1	these hypotheses were partly supported	1

An Example

The example below is laid out as if it were in a scientific publication, but here we are just referring to the hypothesis relevant pieces (… refers to additional context or information the authors would provide).

Abstract: …The primary hypothesis of this study was that the within-player difference in the average of three maximal countermovement jump heights measured 24-hours after a soccer match will be more than 3cm greater following 30 minutes of passive blood-flow restriction (BFR) than following 30 minutes of passive rest administered immediately following a soccer match, in semi-professional soccer players…

Introduction: …The primary aim of the study was to examine the effect of 30 minutes of passive BFR, compared with 30 minutes of passive rest, on recovery immediately following a soccer match in male semi-professional soccer players. Recovery is operationalised here as the average of three maximal countermovement jump heights (cm), measured using a force plate, 24-hours after the soccer match (See the **Table** below for study operations according to PICO and the estimands framework. Different aspects of PICO and the estimands framework should ideally be clear from the introduction or methods section or both sections). The substantive and statistical hypotheses corresponding to this primary aim can be seen below as 'Hypothesis 1'. Our secondary aim was to examine the effect of passive BFR, compared with passive rest, on Borg 6-20 rating of perceived exertion (RPE) taken 24-hours after the soccer match. We did not specify any hypotheses regarding this secondary aim and consider these observations exploratory.

PICO	Estimand Framework	Study Operations
Population	Population	Semi-professional soccer players: male, aged 18-40 who are registered with a league of ireland premier division team for 1 year and had just played 90 minutes of a soccer match.
Intervention	Treatment	BFR: bilateral pneumatic cuffs applied to the proximal thighs at 60% of each participant's individually determined arterial occlusion pressure, for 30 minutes immediately following a soccer match.
Comparator		Sitting comfortably at rest for 30 minutes with no distractions.
Outcome	Variable (or endpoint)	Variable: maximal countermovement jump heights (cm). Endpoint: average of three maximal countermovement jump heights (cm), measured using a force plate, 24-hours after the soccer match.
	Population-Level Summary measure	Mean within-player difference between BFR and rest conditions.
	Intercurrent events	Events and their handling strategy should be explicitly clear from the methods section.

Hypothesis 1:[1] The hypothesis of this study was that the within-player difference in the average of three maximal countermovement jump heights measured 24-hours after a soccer match will be more than 3cm greater following 30 minutes of passive BFR than following 30 minutes of passive rest administered immediately following a soccer match, in male semi-professional soccer players.

[1]The location of the aims will typically be in the introduction, but there is no set location for the substantive or statistical hypotheses or their operationalisations.

The 3cm threshold was specified *a priori* based on a quantitative prediction derived from the theoretical model motivating the study…

$H1_0$: The population-level mean within-player difference (Δ) in the average of three maximal countermovement jump heights measured 24-hours after a soccer match following 30 minutes of passive BFR compared with 30 minutes of passive rest administered immediately following a soccer match will not exceed 3cm.

Formally that is:

$H1_0$: $\Delta \leq 3$

$H1_a$: The population-level mean within-player difference (Δ) in the average of three maximal countermovement jump heights measured 24-hours after a soccer match following 30 minutes of passive BFR compared with 30 minutes of passive rest administered immediately following a soccer match will exceed 3cm.

Formally that is:

$H1_a$: $\Delta > 3$

Discussion: For our claim, there are two possibilities:

1) If the one-sided test of **$H1_0$:** $\Delta \leq 3$cm against **$H1_a$:** $\Delta > 3$cm is statistically significant at the prespecified α level, we can act as if the null hypothesis (**$H1_0$**) is false, in favor of **$H1_a$**. We could therefore conclude that the population-level mean within-player difference in the average of three maximal countermovement jump heights following passive BFR compared with passive rest exceeds the prespecified practically meaningful threshold of 3cm.

2) If the one-sided test is not statistically significant at the prespecified α level, we can conclude that the data do not allow us to reject the null hypothesis (**$H1_0$**) and thus we cannot conclude in favour of **$H1_a$**. This decision applies even if the sample estimate is positive, statistically distinguishable from zero, or greater than 3cm; the prespecified criterion for concluding that the population-level mean within-player difference exceeds 3cm has not been met. This does not mean an effect is absent. Establishing that the mean difference is sufficiently small to be considered practically equivalent to zero would require an appropriately specified equivalence test with prespecified equivalence bounds.